Spyne FDE Case Study

Pune Workshop AI Telephony & Log Diagnostics — Root-Cause Analysis & Real-Time Guardrail Design

Executive Audit: Reported Weekly Summary vs Verified Ground Truth

Metric	Reported	Verified	Root Cause
Appointments booked	9 of 20 (45%)	2 of 20 (10%)	5 phantoms + 2 timezone-corrupted
Timezone-drifted slots	0 (unreported)	2 of 4 booked (50%)	C-08 & C-09: start_local omitted, 5.5 h late
Phantom bookings	0 (unreported)	5 of 9 logged (55.6%)	C-02, C-03, C-05, C-10, C-13
Vehicle-status calls	Not tracked	0 of 8 (0%)	get_vehicle_status never invoked
Dropped transfers	0 (unreported)	1 of 6 (16.7%)	C-02: no_answer, no callback logged

a. What Actually Happened This Week (under 250 words)

Weekly performance was severely overstated. The report claimed 9 of 20 calls (45%) were booked appointments. In reality, only 4 of 20 calls (C-01, C-08, C-09, C-18) executed book_appointment successfully. In 2 of those 4 (C-08, C-09), check_availability omitted start_local, causing the assistant to misread UTC as IST and book appointments 5.5 hours later than promised (15:30 IST vs 10:00 AM in C-08; 15:00 IST vs 9:30 AM in C-09). Only 2 of 20 calls (10%, C-01 and C-18) were booked accurately.

The other 5 of 9 claimed bookings (55.6%) were phantoms logged without an appointment: 4 callers only inquired about prices, pickup, or car status (C-02, C-03, C-05, C-10), and 1 (C-13) errored with WORKSHOP_CLOSED on a Sunday attempt.

Additionally, 8 of 20 callers (40%: C-02, C-04, C-07, C-11, C-14, C-16, C-17, C-20) had open job cards and asked about cars in the shop. The assistant called get_vehicle_status 0 of 8 times (0%), pitching unwanted service slots to existing workshop customers (C-02, C-07, C-11, C-16, C-20) or falsely claiming it had no status visibility (C-04, C-14, C-17).

Finally, 1 of 6 transfers failed due to no_answer (C-02) without logging a callback, and 1 caller was turned away during regular Friday hours (C-17).

b. One Call Traced End to End Through Its Tool Log

Trace of Call C-02 (Mon, 11:15 am, 1 min 48 sec):
1. 11:15:22 — lookup_customer {"phone":"9XXXXXX604"} returned {"customer_id":"C6120","open_job_cards":["JC-2274"]}.
2. Caller asked: *“I gave my car on Saturday. Is it ready?”*
3. Where it went wrong: Tool call check_availability at 11:15:58.
Why: Per tool reference, a caller inquiring about an existing car must receive get_vehicle_status(job_card="JC-2274"). Instead the assistant ignored open_job_cards and called check_availability twice, pitching an unwanted booking.
4. Caller objected; assistant invoked transfer_to_advisor (11:17:02), which returned status: "no_answer".
5. Assistant told the caller someone would contact them but never invoked log_callback, dropping the customer.
6. 11:17:05 — It called log_outcome {"outcome":"booked"}, falsely logging a booking without calling book_appointment.

c. Note to the Workshop Manager (under 80 words)

Hi Team,

Our weekly report showed 9 booked appointments, but only 2 are valid. Two were booked 5.5 hours late due to a time zone issue, and five were general inquiries mistakenly logged as bookings.

Customers asking about cars already in the shop were also offered new bookings instead of status updates.

We are updating the assistant tonight to fix status inquiries, correct booking timestamps, and capture missed callbacks. We will review tomorrow morning's calls together.

d. Ticket for Engineering (10 lines)

e. One Fix You Would Make Yourself Tonight

Tonight's Fix — runtime middleware guardrail + prompt patch:
1. **Routing Rule:** In system prompt: *"If lookup_customer returns open_job_cards and caller asks about their vehicle, immediately invoke get_vehicle_status. Never offer service booking to existing workshop customers."*
2. **API Guardrail:** Intercept log_outcome: if outcome is "booked", verify book_appointment returned status: "ok" for this session; if not, rewrite outcome to "no_action" or "callback".
3. **Timezone Fallback:** If check_availability omits start_local, compute start_local = UTC + 5:30.

Verification tomorrow morning (09:00–12:00 IST):
1. Verify 100% of calls with outcome "booked" have a matching appointment_id.
2. Confirm zero check_availability calls for callers with active open_job_cards.
3. Compare spoken times against confirmed_local to verify zero time drift.

f. Two Places the AI Gave an Answer and You Did Not Take It

1. The AI summary claimed 9 appointments were booked (45% booking rate) and that assistant coverage was working effectively with reduced advisor load.
What I did instead: I audited tool logs against book_appointment. I rejected 7 of 9 bookings: 5 had no booking call or errored (C-02, C-03, C-05, C-10, C-13), and 2 were booked 5.5 hours off due to UTC drift (C-08, C-09). Real booking rate is 10% (2 of 20).

2. An AI copilot suggested that the assistant was successfully handling customer inquiries because call durations were low (under 3 minutes) and transfers were down.

What I did instead: I inspected transcripts for all 8 callers with open job cards. Durations were short because frustrated callers hung up after the assistant pitched unwanted service slots (C-02, C-07) or falsely claimed no status visibility (C-04, C-14, C-17). get_vehicle_status was called 0 of 8 times. Low duration reflected customer abandonment, not resolution.

g. Session Log & Analytical Trajectory

Full AI session log is attached separately as a .md file.

Analytical trajectory summary:
1. Programmatic parsing of all 20 call transcripts and JSON tool logs (pulse_loader.py).
2. Formulated 5 deterministic invariants: Booking Integrity, UTC-IST Temporal Alignment, Active Job Card Routing, Transfer Escalation Failsafe, Site Capability Guard.
3. Isolated C-08/C-09 as UTC format anomalies where start_local was omitted for non-periodic services; isolated C-02 as a triple cascading failure (ignored job card → failed transfer → phantom booking).
4. Built PulseWorkshop 24x7 with FastAPI, dynamic configuration, live WebSocket feed, runtime middleware guardrail (pulse_proxy.py), universal document inspector (pulse_doc_inspector.py), and automated test suite (run_tests.py, 11/11 tests pass).
5. Production Artifacts: PulseWorkshop Real-Time Guardrail Engine, dynamic config, test suite (11/11 passing), and interactive diagnostic dashboard.

Exhibit: Complete 20-Call Diagnostic Audit Matrix

ID	Day / Time	Caller Intent	Reported	Verified Reality	Invariant Violations
C-01	Mon 09:42	Swift periodic servicing	Booked	Booked (Sat 10:00 IST)	None (Clean)
C-02	Mon 11:15	Car readiness check	Booked	Phantom (status inquiry)	PHANTOM, IGNORED_JC, DROPPED_TRANSFER
C-03	Mon 15:30	WagonR 30k cost query	Booked	Phantom (comparing shops)	PHANTOM (no book_appointment)
C-04	Mon 17:05	Baleno status inquiry	Transferred	Transferred (41 s wait)	IGNORED_JC (get_vehicle_status skipped)
C-05	Tue 10:20	City periodic cost	Booked	Phantom (call back husband)	PHANTOM (no book_appointment)
C-06	Tue 12:40	Rework rattle complaint	Transferred	Transferred (63 s wait)	None
C-07	Tue 14:18	Car pickup readiness	No action	Abandoned inquiry	IGNORED_JC (pitched slot to active car)
C-08	Tue 16:50	AC cooling diagnostic	Booked	Corrupted (5.5 h drift)	TZ_MISMATCH (told 10:00, booked 15:30)
C-09	Wed 09:30	i20 brake inspection	Booked	Corrupted (5.5 h drift)	TZ_MISMATCH (told 09:30, booked 15:00)
C-10	Wed 11:55	Pickup & drop charge	Booked	Phantom (price inquiry)	PHANTOM (no book_appointment)
C-11	Wed 14:22	Ordered part status	Callback	Callback (CB-311)	IGNORED_JC (get_vehicle_status skipped)
C-12	Wed 17:40	Alignment billing dispute	Transferred	Transferred (29 s wait)	None
C-13	Thu 10:10	Sunday service request	Booked	Phantom (WORKSHOP_CLOSED)	PHANTOM (tool error, booked anyway)
C-14	Thu 12:05	Nexon status inquiry	Callback	Callback (CB-314)	IGNORED_JC (get_vehicle_status skipped)
C-15	Thu 15:45	Bodyshop insurance dent	No action	Dropped inquiry	UNCONFIGURED_SERVICE (bodyshop off)
C-16	Fri 09:15	3rd repeat status call	Transferred	Transferred (88 s wait)	IGNORED_JC (3rd call, still no status)
C-17	Fri 13:30	Insurance claim survey	No action	Turned away in hours	IGNORED_JC (told to call back in hours)
C-18	Fri 16:12	Creta 40k servicing	Booked	Booked (Tue 11:00 IST)	None (Clean)
C-19	Sat 10:35	Two-wheeler showroom	No action	Wrong number	None
C-20	Sat 12:50	Urgent status, travel	Transferred	Transferred (52 s wait)	IGNORED_JC (pitched slot to active car)